A computer-assisted sharp diagonal Hlawka inequality for real exponents $p \geq 21$

Yuxuan Xu*
School of Computer Science, Wuhan University

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Abstract

We give a computer-assisted proof that, for every real $p \geq 21$, the dimension-independent sharp Hlawka constant of complex ℓ_p is the maximum of an explicit one-parameter ratio on the cyclic triple $(-t,1,1),(1,-t,1),(1,1,-t)$, $1/2 \leq t \leq 2$. The new finite calculation covers $21 \leq p \leq 32$; the range $p \geq 32$ uses a previously accepted Prove2Me theorem. The improvement uses norm caps, clipped pair-coordinate tails, and linear coordinate-sum constraints on a convex averaging domain, together with a sharper residual Hessian estimate. Rational linear-programming dual witnesses and interval certificates cover real exponent intervals, rather than sampled exponents. The finite obligations pass both Arb and a standard-library dyadic rational implementation. We give the analytic reduction, certificate specification, and reproduction instructions. An accompanying Lean formalization of the original uniform complex least-constant goal is verified on Prove2Me with no open dependencies, reusing the accepted $p \geq 32$ tail. Independent analytic review remains pending, and no optimality of the exponent threshold 21 is asserted.

Keywords: Hlawka inequality; ℓ_p norm; sharp constant; computer-assisted proof; interval arithmetic; convexity.

1 Introduction and statement

For $p > 1$ and $v \in \mathbb{C}^n$ write $N_p(v) = (\sum_i |v_i|^p)^{1/p}$. For a triple (x, y, z) put

$$\begin{aligned} S &= N_p(x) + N_p(y) + N_p(z), & T &= N_p(x+y+z), \\ P &= N_p(x+y) + N_p(x+z) + N_p(y+z), & G &= 2S - P. \end{aligned} \tag{1}$$

The Hlawka constant problem asks for the least K, uniformly in the dimension, such that $S - T \leq KG$. Equivalently, the deficit

$$F_{p,K} = (2K-1)S + T - KP = KG - (S - T) \tag{2}$$

must be nonnegative. The triangle inequality gives $G \geq 0$. For diagonal complex matrices this is exactly the corresponding Schatten p-norm question, since their singular values are the absolute values of their diagonal entries. We do not determine the sharp constant for general noncommuting matrices.

Audenaert and Kittaneh formulate a dimension-independent operator version in Section 8.2, Problem 7 of [1]. A survey of the classical inequality and related questions is [2]. The

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diagonal construction and its reduction to a cyclic family were developed in the formalization project [3]. The accepted cutoff-32 result [4, 5] supplies the unbounded tail used here. Our contribution is an explicit certified extension to $[21, 32]$, retaining additional convex constraints in the localization argument. We reproduce the analytic reductions needed for this interval below rather than requiring the reader to combine the earlier working notes.

Define, for $1/2 \le t \le 2$,

$$A_p(t) = (t^p + 2)^{1/p}, \qquad B_p(t) = (2|1-t|^p + 2^p)^{1/p},$$
$$R_p(t) = \frac{3A_p(t) - 3^{1/p}|2-t|}{6A_p(t) - 3B_p(t)}, \qquad K_p = \max_{1/2 \le t \le 2} R_p(t). \tag{3}$$

Each pair of vectors in the cyclic family is nonproportional. Strict convexity of ℓ_p therefore makes the denominator positive. Thus the maximum exists by continuity and compactness. Since $R_p(2) = 1$, $K_p \ge 1$.

Theorem 1.1. *For every real $p \ge 21$, every $n \ge 1$, and every $x, y, z \in \mathbb{C}^n$,*

$$S - T \le K_p(2S - P). \tag{4}$$

The constant K_p is the least constant valid uniformly in n. For each fixed $n \ge 3$, it is also the least constant in that dimension.

The proof on $[21, 32]$ consists of the analytic arguments below and the finite certificate lemma in Section 8. The proof on $[32, \infty)$ invokes [5], whose downloaded source and acceptance metadata are preserved in the repository. That source was not independently rebuilt in this project. The original uniform complex cutoff-21 goal has now also been verified on Prove2Me with zero open dependencies; the formal verification record is described in Section 8.1.

2 Dimension reduction and the scalar envelope

We first explain why it suffices to exclude real three-dimensional counterexamples. Throughout, $p > 1$ and $K \ge 1$.

Lemma 2.1. *If $F_{p,K} \ge 0$ for all triples in $\mathbb{R}^3$, it holds in every finite real dimension and then in every finite complex dimension.*

Proof. For a real triple remove coordinates in which all three entries vanish. Give each remaining coordinate a nonnegative weight w_i, preserving the three pair moments

$$\sum_i w_i\big(|x_i + y_i|^p, |x_i + z_i|^p, |y_i + z_i|^p\big) = \sum_i \big(|x_i + y_i|^p, |x_i + z_i|^p, |y_i + z_i|^p\big).$$

The negative part $-KP$ is constant, and the positive part of $F_{p,K}$ is concave in w, because $s^{1/p}$ is concave and $2K - 1 \ge 0$. If more than three weights are positive, the corresponding moment columns are linearly dependent. Each column is nonnegative and nonzero: three zero pair entries would imply $x_i = y_i = z_i = 0$. Any nonzero kernel direction therefore has both signs. Its feasible interval has two finite endpoints, each with fewer positive weights. Concavity implies that at least one endpoint does not increase $F_{p,K}$. Starting from all weights equal to one and iterating leaves at most three positive weights. Multiply coordinate i by $w_i^{1/p}$ and delete zero coordinates. A counterexample would thus give a counterexample in $\mathbb{R}^3$, padding with zeros when necessary.

For complex vectors put $a_p = (2\pi)^{-1}\int_0^{2\pi} |\cos\theta|^p, d\theta > 0$. For positive integers M, define the real-linear map

$$(Q_M x)_{m,i} = M^{-1/p}\,\mathrm{Re}(e^{2\pi i m/M} x_i), \qquad 0 \le m < M.$$

Riemann-sum convergence and rotation invariance give $N_p(Q_M v) \longrightarrow a_p^{1/p} N_p(v)$ for each of the seven vectors in (1). Apply the real inequality in dimension Mn, take the limit, and divide by $a_p^{1/p}$. This uses convergence of norms, not interchange of a norm with an integral. □

Lemma 2.2 (Weighted three-point convexity)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be convex, let $a_i > 0$, $s = \sum_{i=1}^3 a_i$, and $m = s^{-1}\sum_i a_i u_i$. Then*

$$\sum_{i<j}(a_i + a_j) f\left(\frac{a_i u_i + a_j u_j}{a_i + a_j}\right) \le \sum_i a_i f(u_i) + s f(m).$$

Proof. Order $u_1 \le u_2 \le u_3$; the case of three equal points is immediate. If $u_2 \le m$, the means for pairs $(1,3)$ and $(2,3)$ lie in $[m, u_3]$. Their total mass is $s + a_3$ and their first moment is $sm + a_3 u_3$. The chord bound on this interval bounds their combined contribution by $sf(m) + a_3 f(u_3)$. Jensen bounds the remaining pair by $a_1 f(u_1) + a_2 f(u_2)$. If $m \le u_2$, use the chord on $[u_1, m]$ for pairs $(1,2)$ and $(1,3)$ and Jensen for $(2,3)$. □

Set

$$g_p(u) = ((1+u^p)/2)^{1/p}, \qquad E_p(u) = \frac{1-u}{2(1-g_p(u))} \quad (0 \le u < 1). \tag{5}$$

For a real triple normalized by $S = 1$ and having no zero vector, Lemma 2.2, applied coordinatewise to $f(s) = |s|^p$ with weights $N_p(x), N_p(y), N_p(z)$, gives

$$\sum_{i<j} \frac{N_p(x_i + x_j)^p}{(N_p(x_i) + N_p(x_j))^{p-1}} \le 1 + T^p.$$

Jensen with weights $(N_p(x_i)+N_p(x_j))/2$ implies $P \le 2g_p(T)$. Here x_1, x_2, x_3 denote the three vectors, not scalar coordinates. The function g_p is convex and $g_p(1) = 1$, so its secant slope $(1-g_p(u))/(1-u)$ is nondecreasing; E_p is nonincreasing. A zero vector gives $F_{p,K} = (K-1)$ times the remaining pair gap and cannot be a counterexample. For a counterexample $T < 1$, and

$$K < \frac{1-T}{G} \le E_p(T). \tag{6}$$

Consequently, $E_p(q) < K_p$ forces $T < q$ when $S = 1$.

3 Normalization, orientation, and coarse localization

Fix one of the two windows in Table 1; write its endpoints as p_0, p_1, and put $A = 3q$, $k_0 = cp_0$. All terminating decimals in this paper and the input profiles denote exact rationals. The certificates establish

$$cp < K_p \le p/2, \qquad E_p(q) < K_p. \tag{7}$$

Their verification is specified in Section 8.

The reflection $(x, y, z) \mapsto (-(x+y+z), y, z)$ exchanges T and $N_p(x)$, permutes the pair norms, and changes $F_{p,K}$ by $(2K-2)(T - N_p(x))$. Reflect in a largest singleton norm if

Parameter	Window I	Window II
$[p_0, p_1]$	$[21, 26]$	$[26, 32]$
c	$273/625$	$443/1000$
q	$1011/2500$	$197/500$
Orientation error e	$13/100$	$1/10$
Initial tail error ε_0	10^{-5}	10^{-5}
Refined tail error ε	10^{-8}	10^{-8}
Coarse (ℓ_0, u_0, v_0, w_0)	$(.378, 1.214, .698, 1.214)$	$(.477, 1.183, .750, 1.183)$
Young parameters (e_1, e_2)	$(1, 1/4)$	$(1/10, 1/10)$
Slab width	$1/32$	$1/16$

Table 1: Exact profiles used by the finite verification.

necessary. A counterexample can therefore be chosen with each singleton norm at most T. Scale to $S = 3$. Equations (6)–(7) give, with $N_j = N_p(x_j)$,

$$1 \le T < A, \quad N_j \le T, \quad N_j > 3 - 2A, \quad G < \frac{2}{cp}, \quad k_0 G + T < 3. \tag{8}$$

In three real coordinates each pair has a signed coordinate at least $3^{-1/p}$ times its norm. Since its norm is at most $2A$, its loss is less than e provided

$$2A(1 - 3^{-1/p_0}) < e, \quad 3 - 2A > 2/k_0 + e, \quad 3 - 4/k_0 - 2e > A. \tag{9}$$

These comparisons are certified. The three selected coordinates are distinct. Indeed, if pairs (a, b) and (a, d) select one coordinate with the same sign, adding their inequalities and using the singleton bound for a gives a signed total coordinate at least $3 - 2G - 2e > T$. If they select opposite signs, each inequality, after bounding the other singleton coordinate, gives $\pm a_i \ge N_a - G - e$. Hence $N_a \le G + e < 3 - 2A$, again impossible.

Permute and flip coordinates so that the pair complementary to index i has its positive main coordinate at i. Write X_{ji} for coordinate i of vector j, $P_i = N_p(X_j + X_k)$ and $L_i = X_{ji} + X_{ki}$, where $\{i, j, k\} = \{1, 2, 3\}$. The following rational linear system contains the resulting counterexample:

$$\begin{aligned} 0 \le N_j \le T, \quad \sum_j N_j = 3, \quad 1 \le T \le A, \\ -N_j \le X_{ji} \le N_j, \quad -T \le \sum_j X_{ji} \le T, \\ 0 \le P_i \le N_j + N_k, \quad P_i - L_i \le e, \quad T - k_0 \sum_i P_i \le 3 - 6k_0. \end{aligned} \tag{10}$$

Eighteen rational dual witnesses imply $0 \le L_i \le 3$ and $|(X_j + X_k)_r| \le L_i/2$ for $r \ne i$. Concavity of $s^{1/p}$ yields

$$P_i \le L_i(1 + 2\,2^{-p})^{1/p} \le L_i + 6\,2^{-p}/p < L_i + \varepsilon_0.$$

The last bound is checked at p_0. Add these three linear inequalities to (10). Eighteen further dual witnesses imply the coarse box

$$-u_0 \le X_{ii} \le -\ell_0, \qquad v_0 \le X_{ji} \le w_0 \quad (j \ne i). \tag{11}$$

For clarity, a dual witness for $Av \le b$, $Ev = f$ consists of rational $y \le 0$, z with $A^T y + E^T z = d$. It proves $d^T v \ge b^T y + f^T z$. The verifier reconstructs all objectives and checks these identities exactly; an optimizer's claimed optimum is never used as evidence.

3.1 A smaller tail bound

In any box with parameters (ℓ, u, v, w) as in (11), a non-main coordinate of the pair excluding k equals $V_i - X_{ki}$, where $V = \sum_j X_j$ and $i \neq k$. Since $X_{ki} \geq v$ and $V_i \leq N_p(V) \leq A$, it lies in

$$[v - u,\ \min(w - \ell, A - v)]. \tag{12}$$

Window I uses this clipped interval. Its coarse tail bound is .516, and its main coordinate is at least 1.396, giving ratio 129/349. Window II uses the box-only tail bound .706 and main bound 1.5. In both windows the certificate checks, with the corresponding ratio r_0,

$$\frac{6}{p_0} r_0^{p_0} < \varepsilon, \qquad 0 < r_0 < 1. \tag{13}$$

Thus $P_i \leq L_i + \varepsilon$ for all normalized counterexamples. The clipping argument uses the known total norm cap; it makes no claim about arbitrary points of the coarse box without that cap.

4 Certified contraction

Let $a_i = -X_{ii}$, $w_i = \sum_{r \neq i} X_{ir}$, $W = \sum_i w_i = \sum_i L_i$, and $\gamma_p = 2\,2^{-1/p}$. Hölder gives

$$a_i^p + (w_i/\gamma_p)^p \leq N_i^p, \qquad w_i \leq \gamma_p N_i. \tag{14}$$

From $P = 6 - G$ and the refined tails, $W > 6 - 2/(cp) - 3\varepsilon$. On each rational slab $[\alpha, \beta]$ choose certified rational bounds

$$\begin{aligned} &t_- \leq 2^{-1/\alpha} \leq t_\alpha^+, && 2^{-1/\beta} \leq t_\beta^+, \\ &h \geq 2/(c\alpha) + 3\varepsilon, && D \geq 2/(c\alpha) + 3\varepsilon - 6(1 - t_\alpha^+), \\ &D \geq 0, && \delta \geq D/(2t_-), \quad \delta \geq 0. \end{aligned} \tag{15}$$

The comparison $2/c > 6 \log 2$ is certified. It makes $6\,2^{-1/p} + 2/(cp)$ decreasing. Therefore the nonnegative residuals $r_i = \gamma_p N_i - w_i$ satisfy

$$\sum_i r_i < D, \qquad W > 6 - h, \qquad 2t_- \leq \gamma_p \leq 2t_\beta^+. \tag{16}$$

In particular, a pair of rows consumes the same total budget D, not two separate copies of that budget.

Lemma 4.1 (Diagonal and ratio bounds)**.** *Suppose $\delta \leq N \leq V$, $a^p + (w/\gamma_p)^p \leq N^p$, and $w \geq \gamma_p N - D$. If $B \geq 0$ and*

$$(B/V)^p + (1 - \delta/V)^p \geq 1, \tag{17}$$

then $a \leq B$. Also, if $N_i \geq n > 0$ and

$$\begin{aligned} &1 \leq s \leq 2t_- - D/n, && 0 \leq \tau \leq s/2 \leq \sigma, \\ &\tau^p + (s - \tau)^p > 1, && \sigma^p + (s - \sigma)^p > 1, \end{aligned} \tag{18}$$

with all four power bases in $[0, 1]$, then each off-diagonal entry of row i lies strictly between τN_i and σN_i.

Proof. Since $D/\gamma_p \leq \delta$, the first hypotheses imply $a^p \leq N^p - (N - \delta)^p \leq V^p - (V - \delta)^p \leq B^p$; the middle function is increasing for $N \geq \delta$. For the second claim divide the two off-diagonal entries by N_i, obtaining $u, v \geq 0$, $u^p + v^p \leq 1$, $u + v \geq s$. Both are at most one. If $u \leq \tau$, use the decrease of $r^p + (s - r)^p$ on $[0, s/2]$. If $u \geq \sigma$, use its increase on $[s/2, s]$; $u \leq 1 \leq s$. Either case contradicts the stated power comparison. The same applies to v. □

Initialize $n = v_0$, which is a valid lower bound on each N_i. The following updates are applied sequentially, always using the already established lower bound and checking $\delta \leq n$.

Lower update. Choose $n' > 0$ and $B \geq 0$ satisfying (17) at $V = n'$ and $A \leq \tau(3-n') - B$. If $N_i < n'$, the diagonal bound gives $a_i \leq B$, whereas the ith total coordinate exceeds $\tau(3 - N_i) - B > A$, a contradiction. Hence $N_i \geq n'$.

Upper update. Choose $0 < U < 3$ and $B \geq 0$ satisfying (17) at $V = 3 - U$ and

$$2t_\beta^+ A \leq 6 - h - \sigma(3 - U) - B. \tag{19}$$

If $N_i > U$, let $M = N_j + N_k < 3 - U$. Minkowski applied to (14) gives

$$(a_j + a_k)^p + ((w_j + w_k)/\gamma_p)^p \leq M^p.$$

The sum of their residuals is less than D, and $M \geq 2n \geq \delta$. The diagonal bound thus gives $a_j + a_k \leq B$. The sum of total coordinates j, k exceeds $W - \sigma M - B > 6 - h - \sigma(3 - U) - B$. Hölder bounds this sum by $\gamma_p T < 2t_\beta^+ A$, contradicting (19). Thus $N_i \leq U$.

Final box. With the final n, U, τ, σ, choose A_f satisfying the diagonal certificate at $V = U$. Then

$$\tau n < X_{ij} < \sigma U \ (i \neq j), \qquad \tau(3 - U) - A < a_i \leq A_f. \tag{20}$$

Each slab stores a rational box $B = B(\ell, u, v, w)$, verified by

$$v \leq \tau n, \quad \sigma U \leq w, \quad A_f \leq u, \quad \ell \leq \tau(3 - U) - A. \tag{21}$$

There are ten lower updates followed by the final ratio and upper update in every slab. All power comparisons are checked at $p = \beta$ with bases in $[0, 1]$, so they hold throughout the slab. These are finite induction steps, with no assumption of the desired final box in an earlier update.

5 Convex domains retaining the normalization information

The normalization $S = 3$ is useful for deriving bounds but is not itself a convex constraint to preserve during averaging. Window II uses

$$\mathcal{D}_2 = B \cap \{N_p(X_i) \leq U \ (i = 1, 2, 3), \ N_p(V) \leq A\}. \tag{22}$$

For Window I, each pair gap is nonnegative and at most G; hence at a normalized counterexample

$$P_i = 3 - N_i - (N_j + N_k - P_i) > 3 - U - 2/(c\alpha).$$

By the tail bound, set

$$J = 3 - U - 2/(c\alpha) - \varepsilon, \qquad W_0 = 6 - h. \tag{23}$$

Then the counterexample belongs to the smaller domain

$$\mathcal{D}_1 = \mathcal{D}_2 \cap \{L_i \geq J \ (i = 1, 2, 3), \ \sum_i L_i \geq W_0\}. \tag{24}$$

Both domains are convex and invariant under simultaneous permutation of rows and coordinates. The new constraints in (24) are linear. In particular, neither domain retains the equality $S = 3$.

Here are the constants used to bound the Hessian. Set $m = \min(\ell, v)$, $M = \max(u, w)$. In Window I set

$$\begin{aligned} L &= \max(2v, J), \quad s = L - u, \quad t = \min(2w - \ell, A), \\ \eta &= \max\{|v - u|, |\min(w - \ell, A - v)|\}, \quad b_0 = \max(v, W_0/6). \end{aligned} \tag{25}$$

The interval in (12) is also checked to be nonempty. In Window II use the weaker box bounds

$$L = 2v, \quad s = 2v - u, \quad t = 2w - \ell, \quad \eta = \max(|v - u|, |w - \ell|), \quad b_0 = v. \tag{26}$$

The certificate verifies $m > 0$, $0 < s \le t$, $0 < \eta \le L$ and, in Window I, $J, W_0 > 0$. On the appropriate domain, each singleton coordinate has absolute value in $[m, M]$, total coordinates lie in $[s, t]$, and each pair has main coordinate at least L and tails of magnitude at most η. Singletons, pairs, and total are therefore nonzero.

6 The Hessian estimate

Write $|\cdot|_2$ for the Euclidean norm. For $p > 2$, N_p is C^2 away from the zero vector, even if some individual coordinates vanish. Differentiation and completion of the square give

$$H_p(v, z) = \frac{p - 1}{N_p(v)^{p-1}} \min_{a \in \mathbb{R}} \sum_i |v_i|^{p-2}(z_i - a v_i)^2. \tag{27}$$

The minimizing a is $\sum_i |v_i|^{p-2} v_i z_i / \sum_i |v_i|^p$. For each row decompose a direction as $Z_j = a_j X_j + U_j$ using this minimizer, and put $D_Z = \sum_j Z_j - bV$ using the total minimizer. The norm caps in the convex domain imply the positive lower bounds

$$\sum_j H_p(X_j, Z_j) \ge b_C \sum_j |U_j|_2^2, \quad H_p(V, \sum_j Z_j) \ge b_T |D_Z|_2^2, \tag{28}$$

where

$$b_C = \frac{(p-1)m^{p-2}}{U^{p-1}}, \qquad b_T = \frac{(p-1)s^{p-2}}{A^{p-1}}, \qquad d = \frac{(p-1)\eta^{p-2}}{L^{p-1}}.$$

In a pair with main coordinate k, choose in (27) the scalar which eliminates the main coordinate. The resulting upper bound is d times the sum of squares of its two remaining components.

6.1 Controlling the radial coefficients

Let $a_j = \max a_i$, $a_k = \min a_i$, $\Delta = a_j - a_k$, and $R = \sum_i a_i X_i$. Then

$$V_j R_k - V_k R_j \ge m(V_j + V_k)\Delta. \tag{29}$$

Indeed, subtract a_k from all coefficients, and let l be the third index and $\lambda = a_l - a_k \in [0, \Delta]$. The left side becomes $\Delta A_1 + \lambda A_2$, where $A_1 = V_j X_{jk} - V_k X_{jj}$ and $A_2 = V_j X_{lk} - V_k X_{lj}$. If $A_2 \ge 0$ it is at least ΔA_1; otherwise it is at least $\Delta(A_1 + A_2)$. Both A_1 and $A_1 + A_2 = V_k X_{kj} - V_j X_{kk}$ are at least $m(V_j + V_k)$.

Define

$$H = \frac{s^2 + t^2}{m^2(s + t)^2}. \tag{30}$$

For two positive numbers in $[s, t]$, their squared Euclidean norm divided by their squared sum is at most $(s^2 + t^2)/(s + t)^2$; this follows by monotonicity in their ratio. Since $R = D_Z - \sum U_i + bV$, Cauchy and Young's inequality in (29) give

$$\Delta^2 \le H \left((1 + e_1)|D_Z|_2^2 + 3(1 + e_1^{-1}) \sum_i |U_i|_2^2 \right). \tag{31}$$

6.2 Pair residuals

For the pair of rows j, r complementary to k, the residual in coordinate $i \neq k$ equals $r_{ki} + e_{ki}$, where

$$r_{ki} = (a_j - a_r)\frac{X_{rk}X_{ji} - X_{jk}X_{ri}}{X_{jk} + X_{rk}},$$
$$e_{ki} = U_{ji} + U_{ri} - \theta_{ki}(U_{jk} + U_{rk}), \qquad \theta_{ki} = \frac{X_{ji} + X_{ri}}{X_{jk} + X_{rk}}. \tag{32}$$

The fraction multiplying $a_j - a_r$ is bounded by M in absolute value: it is a difference with positive weights summing to one, and both relevant coordinates have absolute value at most M. Thus

$$\sum_{k,i\neq k} r_{ki}^2 \leq 2M^2 \sum_{j<r}(a_j - a_r)^2 \leq 4M^2\Delta^2. \tag{33}$$

For Window II, $|\theta_{ki}| \leq 1$ and four-term Cauchy gives $\sum e_{ki}^2 \leq 16 \sum_j |U_j|_2^2$: in the six residuals an off-diagonal entry occurs at most three times and a diagonal entry twice.

For Window I put $\rho = \eta/L$. Split e_{ki} into $U_{ji} + U_{ri}$ and $-\theta_{ki}(U_{jk} + U_{rk})$. In each coordinate, the first set contributes $(x+y)^2 + (x+z)^2 \leq 3(x^2+y^2+z^2)$, since the difference is $(x-y-z)^2 + (y-z)^2$. The second set has squared norm at most $4\rho^2 \sum |U_j|_2^2$, because each off-diagonal main entry occurs twice. The Euclidean triangle inequality yields

$$\sum e_{ki}^2 \leq (\sqrt{3} + 2\rho)^2 \sum_j |U_j|_2^2 \leq E \sum_j |U_j|_2^2, \qquad E = (7/4 + 2\rho)^2. \tag{34}$$

For Window II set $E = 16$. Using Young's inequality with e_2, (31) and (33) give

$$\sum (r_{ki} + e_{ki})^2 \leq G_T |D_Z|_2^2 + G_C \sum_j |U_j|_2^2, \tag{35}$$
$$G_T = 4M^2(1 + e_2)H(1 + e_1),$$
$$G_C = 12M^2(1 + e_2)H(1 + e_1^{-1}) + E(1 + e_2^{-1}).$$

Combining the positive and negative Hessian bounds proves

$$D^2F_{p,K}(X)[Z,Z] \geq (b_T - KdG_T)|D_Z|_2^2 + ((2K-1)b_C - KdG_C)\sum_j |U_j|_2^2. \tag{36}$$

Put

$$r_T = \frac{A}{L}\left(\frac{\eta A}{Ls}\right)^{p-2} = \frac{d}{b_T}, \qquad r_C = \frac{U}{L}\left(\frac{\eta U}{Lm}\right)^{p-2} = \frac{d}{b_C}. \tag{37}$$

On every slab the certificate checks the two strict inequalities

$$1 - G_T(p/2)r_T > 0, \qquad 2c - 1/p - (G_C/2)r_C > 0. \tag{38}$$

By (7), both coefficients in (36) are positive. In particular $F_{p,K}$ is convex on the relevant domain: along each segment it is C^2 in a neighborhood of every point and has nonnegative second derivative. This also applies to segments on the boundary of the closed domain.

7 Permutation averaging and sharpness

Average X over all six simultaneous row and coordinate permutations. The resulting matrix has diagonal entries $-a$ and off-diagonal entries $b > 0$. It remains in the convex domain,

and convexity gives $F_{p,K}(\overline{X}) \le F_{p,K}(X)$. The box gives $a \le u$, $b \ge v$; Window I also gives $b \ge W_0/6$, hence $b \ge b_0$ in both windows. The certificate verifies

$$u/v \le 2, \qquad \ell/w \ge 1/2 \quad \text{or} \quad A \le 3b_0/2. \tag{39}$$

If the box alternative holds, $a/b \ge 1/2$ directly. Otherwise the positive total coordinates give $0 < 2b - a \le A$, and therefore $a/b \ge 2 - A/b \ge 1/2$. Thus $t = a/b \in [1/2, 2]$. For $K = K_p$, equation (3) implies

$$F_{p,K}(\overline{X}) = b(6A_p(t) - 3B_p(t))(K_p - R_p(t)) \ge 0.$$

A counterexample would instead give $F_{p,K}(\overline{X}) \le F_{p,K}(X) < 0$. This excludes real three-dimensional counterexamples on both windows. Lemma 2.1 proves the complex inequality on $[21, 32]$. The accepted theorem [5] supplies $p \ge 32$ with the same definitions, proving admissibility in Theorem 1.1.

Finally, the cyclic triple has strictly positive G and ratio exactly $R_p(t)$. A constant valid in complex dimension three must dominate every such ratio, and hence K_p. Taking a maximizer shows attainment. Padding with zeros gives the same sharpness in each dimension $n \ge 3$.

8 Finite certificates and reproducibility

Lemma 8.1 (Finite certificate lemma)**.** *For the two profiles in Table 1, all scalar, orientation, linear-dual, tail, contraction, geometry, orbit, and curvature comparisons used in Sections 3–7 hold on their specified closed real exponent intervals.*

Computer-assisted verification. The proof is the deterministic replay described here, using the committed JSON witnesses and the directed interval operations of Appendix A. Table 2 records the finite size. All profile parameters, partition endpoints, and witnesses are parsed as exact fractions. Partitions must meet exactly, in order, with positive length and matching outer endpoints. Thus all real exponents are covered.

For each scalar cell $[a, b]$, the witness supplies one rational $t \in [1/2, 2]$. Evaluate $R_p(t) - E_p(q)$ and $R_p(t) - cp$ with $p \in [a, b]$ and require positive interval lower endpoints. This proves the lower bounds in (7). For the upper bound, partition the exponent range and, on each exponent cell, partition the entire t range. Evaluate $p/2 - R_p(t)$ on every rectangle and require a positive lower endpoint. Every denominator is checked positive before division.

The coarse checker reconstructs (10) and the 36 objectives; it checks each rational dual identity, sign and target, adding the initial tail constraints only between its two stages. The scalar orientation, tail and monotonicity prerequisites are checked independently. For each contraction slab it validates (15), the preceding n in each round, the bases and branch conditions in Lemma 4.1, the updates, and (21). It then reconstructs all geometric constants as rational numbers and checks their domains and (39). Finally, interval evaluation of (37)–(38) covers the full slab. Stored floating-point diagnostics are not used in any acceptance decision. □

The primary files are `profiles/cutoff21.json`, `profiles/cutoff26.json` and the corresponding subdirectories of `certificates/`, each containing `scalar.json`, `cyclic-upper.json`, `coarse.json`, and `contractor.json`. The acceptance implementation is `scripts/verify.py`, with exact model construction in `scripts/coarse_lp.py` and geometric constants in `scripts/geometry.py`. Candidate generation may use floating-point optimization, but the replay neither calls that optimizer nor trusts its reported optimum.

Obligation	$[21, 26]$	$[26, 32]$
Scalar exponent cells	2355	2699
Upper-bound rectangles	960	672
Exact LP dual witnesses	36	36
Contraction slabs	160	96
Lower updates per slab	10	10

Table 2: Certificates cover intervals, not isolated sample points.

Both windows have passed Arb at 192 and 256 bits and the standard-library rational backend at 256 bits. The minimum reported lower endpoints of the two dimensionless curvature margins on $[21, 26]$ are approximately 0.391298 and 0.321741; on $[26, 32]$ they exceed 0.3200 and 0.2675. These rounded summaries are descriptive; the actual decisions use the unrounded interval endpoints.

From the repository root, the complete dependency-free release replay is

```
python -S scripts/replay_release.py
```

It runs both windows at 256-bit rational precision, rejection controls, and checks all eleven preserved upstream file hashes. It writes `results/release-replay.json`, including the input hashes. Python 3.14 was used for the recorded runs. No third-party package is imported by this replay; `-S` disables site initialization. For an individual window, for example:

```
python -S scripts/verify.py --backend rational --precision 256 \
  --profile profiles/cutoff21.json \
  --directory certificates/cutoff21 \
  --output results/cutoff21-rational.json
```

The backslash continuation above is for a POSIX shell; on Windows enter the command on one line. An Arb replay uses the same command without `-S`, with `-backend arb`, after installing `python-flint==0.9.0` (see `requirements.txt`).

Negative controls deliberately damage coverage, a cyclic witness, a contraction chain, an LP dual, a certificate set, a final box, and a curvature configuration. Window I additionally tests an unjustifiably small clipped-tail budget. Both backends reject all eight Window I mutations; Window II has seven. There are also 53 arithmetic cross-checks against higher-precision Arb. Rejection tests are implementation checks, not substitutes for the analytic lemmas.

8.1 Trust and provenance

The two arithmetic backends share the certificate data, LP model, geometric formulas, and `verify.py`. They are not independent end-to-end proofs. Their numerical checks rely on Python's integer arithmetic, interpreter, interval implementation, and the correspondence between the mathematical obligations and the checker.

A separate Lean formalization verifies the new $[21, 32]$ argument and the passage to arbitrary finite complex dimensions, including dimension zero, zero vectors and unequal singleton norms. The original theorem `HlawkaSchatten.DiagonalConstruction.campaign_cutoff21` is `Proved` on Prove2Me with zero open dependencies: https://prove2.me/theorems/db286144-c8c4-4918-9bc9-03fcadedfffd. The new local proof chain is checked with Lean 4.33.1 and Mathlib commit `0df444a360eaa60ab8c11dca51a86af692955474`; its axiom checks use only `propext`, `Classical.choice` and `Quot.sound`. The final platform

proof imports the already proved cutoff-32 tail, whose full proof has not been independently rebuilt here. Source hashes, exact certificate checks and platform read-backs accompany the code. These formal checks do not constitute independent analytic review of the manuscript, which remains pending.

The tail reference is the final `theorem solution` in submission `2d248eb1-4938-4dee-8392-df24d2c70f2f`, with statement "for every real $p \geq 32$, the cyclic constant is the least uniform complex ℓ_p Hlawka constant." Its record and byte-preserved source are in `sources/`; `sources/provenance.json` gives identifiers and SHA-256 digests. The coarse, contraction and convexity components at cutoff 32 are attributed in those records to `@bwdaugherty`. The underlying public construction repository is [3], with reference commit `1829591140522193a81ae721330e4d775c81123e`. We do not count a source download or acceptance record as a local rebuild.

9 Discussion

The number 21 is a sufficient lower threshold for the exponent p, not the value of K_p and not a proven optimal threshold. The constant itself remains the exact real maximum (3); a decimal approximation to it would require a specified exponent and precision. The proof does not require p or the cutoff to be an integer. The older window $[127/4, 32]$ illustrates that rational cutoffs work in the same certificate framework.

The gain comes from keeping information during convexification: singleton and total norm caps reduce Hessian denominators; the total norm clips pair tails; pair-main-coordinate floors improve the radial scale; and the off-diagonal sum floor keeps the permutation average in the cyclic parameter range. These constraints are convex or linear, which is essential for Jensen averaging. Further threshold reductions would require new certificates and, if estimates change, a corresponding analytic justification. No claim for $2 < p < 21$ follows from the present calculation.

A Directed dyadic arithmetic

At precision B an integer pair (l, u) represents $[l2^{-B}, u2^{-B}]$. Fraction inputs are enclosed using integer floor and ceiling. Addition and subtraction use endpoint identities. Multiplication and division use all endpoint combinations and outward integer rounding; division rejects intervals containing zero. Float inputs are rejected. Strict positivity is accepted only when the integer lower endpoint is strictly positive. These rules give containment by elementary interval arithmetic, regardless of dependency-induced overestimation.

For a positive dyadic x, range reduction writes $x = 2^k m$, with $1 \leq m < 2$. Let $z = (m-1)/(m+1)$; the implementation checks a rounded enclosure in $[0, 1/2]$. The finite formula

$$\log m = 2\sum_{j=0}^{191} \frac{z^{2j+1}}{2j+1} + R, \qquad 0 \leq R \leq \frac{2z^{385}}{385(1-z^2)} \leq \frac{8}{3 \cdot 385 \cdot 2^{385}}$$

follows by integrating the geometric series or the power series for $\operatorname{arctanh} z$. All terms are evaluated outward, and the rational remainder is added to the upper endpoint. Compute $\log 2$ by the same formula and use $\log x = k \log 2 + \log m$, with signed interval multiplication when $k < 0$.

For e^x, choose $k \geq 0$ so that the interval for $z = x/2^k$ lies in $[-1/2, 1/2]$, and evaluate

$$e^z = \sum_{j=0}^{80} \frac{z^j}{j!} + R, \qquad |R| \leq \frac{2}{2^{81} 81!}.$$

The Lagrange remainder and $e^{1/2} < 2$ justify this bound. Enlarge both endpoints by the remainder bound, verify positivity, and square k times outward. Monotonicity gives interval exponential and logarithm from their endpoint enclosures. For positive bases use $x^y = \exp(y \log x)$. When the base interval includes zero and its upper endpoint is positive, with strictly positive exponents, monotonicity in the base gives lower endpoint zero and the upper-base power enclosure. A zero base gives zero. Other nonpositive-base cases are rejected. This includes the vanishing factor $|1 - t|^p$ at $t = 1$. The implementation is `scripts/rational_intervals.py`; explicit remainders, not presumed numerical precision, supply rigor.

Acknowledgments and AI-use disclosure

The cyclic reduction and formalized baseline are credited to the `savarin/hlawka-schatten` project (Ezzeri Esa / savarin in the source attribution) and the Prove2Me contributors, including `@bwdaugherty`. Their sources and licenses are identified in `sources/NOTICE.md`. This project adapts that proof architecture; it does not claim to originate the diagonal Hlawka problem or the cyclic extremizer construction.

OpenAI Codex assisted with mathematical exploration, proof drafting, certificate generation, implementation, Lean formalization, verification runs, and preparation of this manuscript. AI assistance and internal replay do not constitute independent expert review. The author name and institutional affiliation are supplied by the author; no institutional endorsement is implied.

Data and software availability

The manuscript source, PDF, exact input certificates, replay scripts, and verification reports are available at https://github.com/Anchen-AI4Math/hlawka-diagonal-certificates. This is a repository preprint, not a journal publication. Citation metadata and third-party notices accompany the release.

References

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